BI Project E-commerce Analysis

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1. Context & Objective

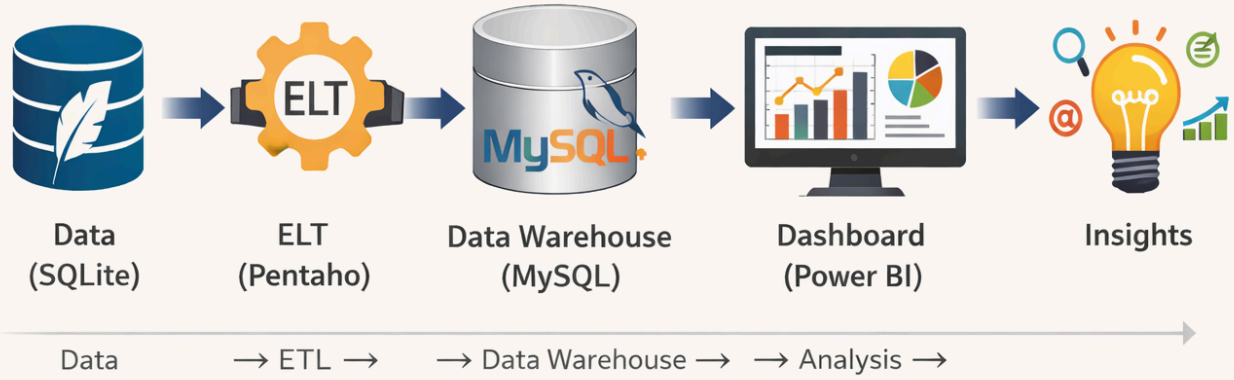
Context

This project uses the Brazilian e-commerce dataset provided by Olist, available on Kaggle, in SQLite format.
The dataset contains information about orders, customers, products, sellers, payments, deliveries, and customer reviews.

Objective

The goal of this project is to build an end-to-end Business Intelligence solution to:

- Design a Data Warehouse (DWH) using a Star Schema
- Define and compute key business KPIs
- Build a multi-page analytical report and an executive dashboard
- Provide actionable insights to support business decision-making



2. Technical Architecture

Technology Stack

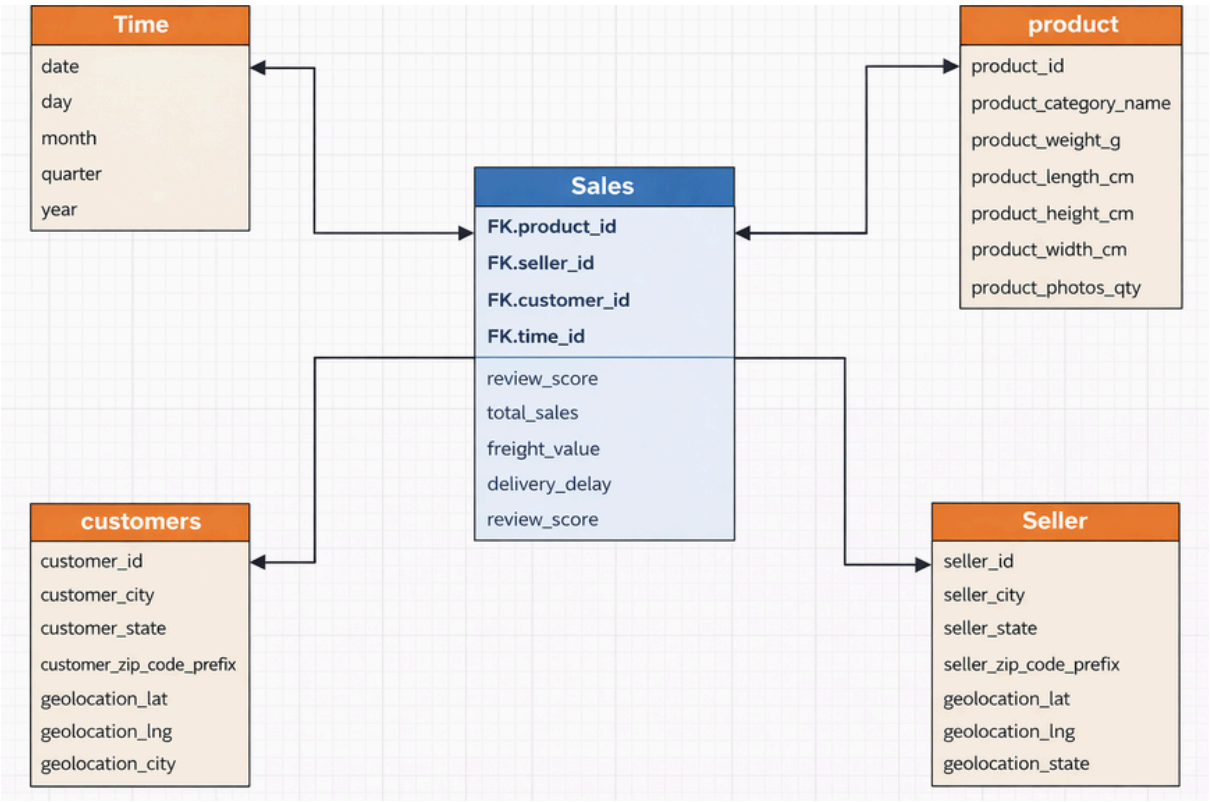
- Data Source: SQLite database (olist.sqlite)
- Data Processing / Transformation: ELT pipeline for data preprocessing.
- Data Warehouse: MySQL (Star Schema)
- BI & Visualization: Power BI

Workflow Overview

- Extract data from SQLite (Olist dataset)
- Transform & preprocess data using ELT (clean missing values, standardize columns, enrich with calculations)
- Load the cleaned data into the MySQL Data Warehouse
- Model the data in Star Schema (fact & dimension tables)
- Connect Power BI to MySQL DWH for building reports and dashboards

Data Modeling

- Fact Table: Orders / Sales
- Dimension Tables: Customers, Products, Categories, Sellers, Time,



Executive Dashboard & KPIs

Reporting Period: January 2017 – December 2018

Data Source: Olist public dataset

Main KPIs

- Total Revenue
- Total Orders
- % of Delivere
- Average Delivery Delay
- Average Customer rating

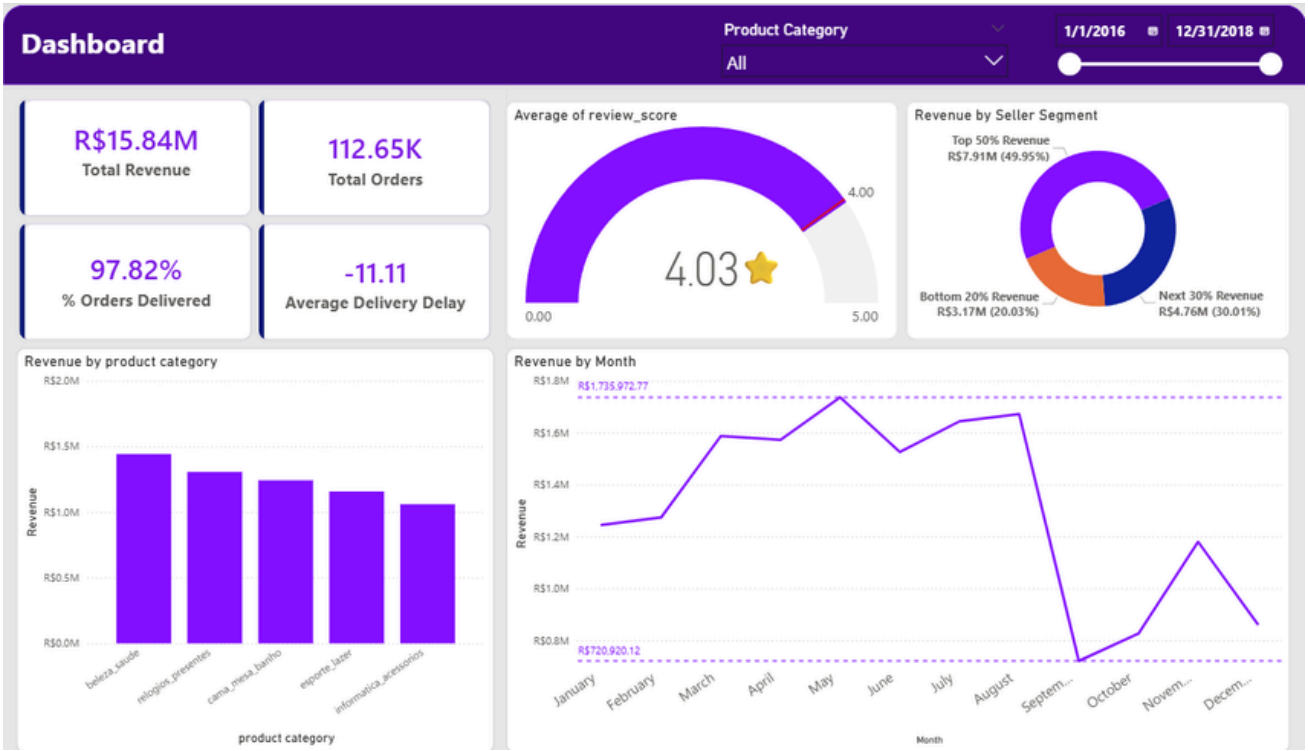


Figure : Executive dashboard summarizing key business KPIs.

3. Analytical Report Pages

Power BI Report Pages

- Overview
- Time Analysis
- Product Analysis
- Delivery & Customer Satisfaction
- Sellers Performance

Enable business analysts to explore performance across:

- Time
- Product categories
- Geographical areas
- Logistics performance
- Seller performance

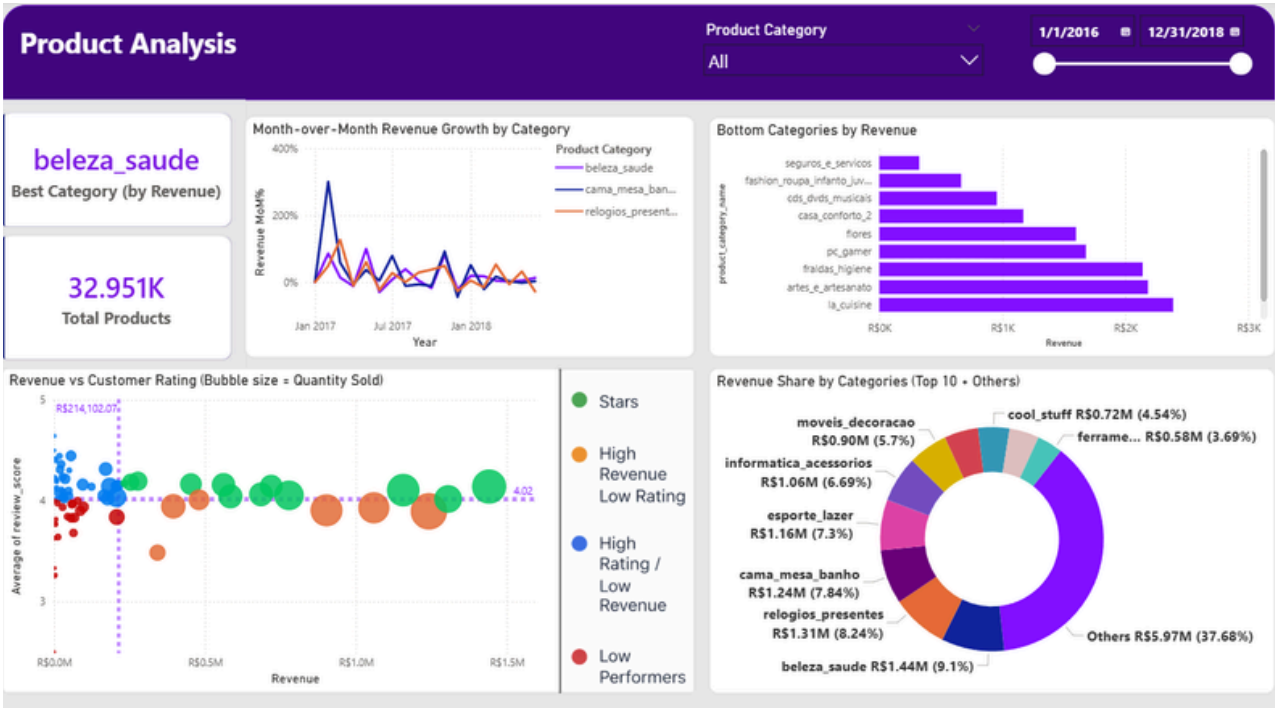


Figure : Product Performance Analysis Page

4. Business Insights

1

Sales Seasonality

We notice a significant decrease in revenue in September compared to the previous months, followed by a stable increase

Business Action:

Perform a detailed analysis of September to understand the reasons behind this revenue decline.

2

Revenue vs. Review Score

High-revenue categories do not always have the highest customer ratings.

Business Action:

improve product quality or post-purchase support for top-selling categories to protect long-term customer satisfaction.

3

Delivery Impact on Satisfaction

Delivery delays negatively impact customer review scores.

Business Action:

prioritize reliable sellers and logistics partners.

4

Seller Performance Concentration

Delivery delays negatively impact customer review scores.

Business Action:

design loyalty or incentive programs for top sellers.

5

Pareto Effect on Products

A small number of product categories generate the majority of revenue.

Business Action:

focus marketing and inventory efforts on high-performing categories.

6

Delivery Performance by State

Delivery delays vary significantly by state.

Business Action:

Review logistics partners in slow-performing states.

5. Conclusion & Skills Demonstrated

Conclusion

This project demonstrates a complete Business Intelligence workflow, from raw data extraction to actionable insights and executive-level dashboards. Analysis of the Olist e-commerce dataset revealed key trends in sales, delivery performance, customer satisfaction, and seller contributions, providing clear recommendations for data-driven business decisions.

Skills Demonstrated

- **Data Warehouse Design**: Building a Star Schema for efficient reporting.
- **Advanced SQL**: Writing complex queries for KPI calculation and data transformation.
- **KPI Definition & Measurement**: Identifying and tracking key performance indicators.
- **Data Visualization & Storytelling**: Developing interactive Power BI dashboards to communicate insights effectively.
- **Business Analysis & Recommendations**: Translating analytical findings into actionable business strategies.